Phase Transition in Embedding-Based Graph Retrieval: A Formal Proof

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1 Setup

Let $G = (V, E)$ be a graph with $|V| = n$ and mean degree $\bar{d}$. Fix a query entity $q \in V$ and an answer entity $a \in V$ at shortest-path distance h from q. Let $\mathcal{B}_h(q) \subseteq V$ denote the h-hop neighborhood of q.

Let $f : V \to \mathbb{R}^D$ be an embedding function mapping each node to a unit vector in $\mathbb{R}^D$. For a query embedding $\mathbf{q} = f(q)$, define the *similarity score* of node v as $S_v = \langle \mathbf{q}, f(v) \rangle$ (cosine similarity, since all embeddings are unit-normalized).

Retrieval task. Given q, retrieve a by selecting $\hat{a} = \arg\max_{v \in \mathcal{B}_h(q)} S_v$.

2 Model Assumptions

A1. (Neighborhood size.) $|\mathcal{B}_h(q)| = M$, where $M \approx \bar{d}^h$ for locally tree-like graphs with mean degree $\bar{d}$ (cf. the Galton-Watson approximation; van der Hofstad, 2017).

A2. (Noise distribution.) For each noise candidate $v \in \mathcal{B}_h(q) \setminus \{a\}$, the similarity score S_v is drawn independently from $\mathcal{N}(0, \sigma^2)$, where $\sigma^2 = 1/D$.

This follows from the concentration of cosine similarity between independent random unit vectors on $\mathbb{S}^{D-1}$: if $\mathbf{q}$ and $f(v)$ are independent uniform random unit vectors in $\mathbb{R}^D$, then $\langle \mathbf{q}, f(v) \rangle$ has mean 0, variance $1/D$, and converges in distribution to $\mathcal{N}(0, 1/D)$ as $D \to \infty$.

A3. (Signal margin.) The answer node has similarity score $S_a \sim \mathcal{N}(\delta, \sigma^2)$ for some *signal margin* $\delta > 0$. That is, the answer's embedding is δ closer to the query in expectation than a random node.

Remark 1. *Assumption A2 is the key modeling choice. In practice, embeddings are not independent uniform random vectors—nodes in the same neighborhood share structural and textual features, inducing positive correlations. However, A2 provides a* lower bound *on retrieval difficulty: correlated noise candidates are* harder *to distinguish from the signal (the effective noise variance increases), so the true h^* is at most the value we derive under A2.*

3 Main Result

Theorem 1 (Phase Transition in Graph Retrieval)**.** *Under assumptions A1–A3, let $\Delta = \delta/\sigma = \delta\sqrt{D}$ denote the normalized signal-to-noise ratio. Then the probability of correct retrieval satisfies:*

$$P(\textit{correct}) = \int_{-\infty}^{\infty} \Phi(x + \Delta)^{M-1} \, \phi(x) \, dx \tag{1}$$

where Φ and ϕ are the standard normal CDF and PDF, and $M = \bar{d}^h$.

Moreover, $P(\text{correct})$ undergoes a phase transition at the critical hop count

$$\boxed{h^* = \frac{\delta^2 D}{2 \log \bar{d}}} \tag{2}$$

in the following sense:

- *If $h < h^* - \omega(1)$, then $P(\text{correct}) \to 1$ as $D \to \infty$.*
- *If $h > h^* + \omega(1)$, then $P(\text{correct}) \to 0$ as $D \to \infty$.*

4 Proof

We prove Theorem 1 in three steps.

Step 1: Exact retrieval probability (the MAFC integral)

Retrieval succeeds when $S_a > \max_{v \neq a} S_v$. Let $Z = (S_a - \delta)/\sigma$ and $W_i = S_{v_i}/\sigma$ for $i = 1, \ldots, M-1$. Then $Z \sim \mathcal{N}(0,1)$ and $W_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$, and:

$$\begin{aligned} P(\text{correct}) &= P\Big(S_a > \max_i S_{v_i}\Big) \\ &= P\Big(\sigma Z + \delta > \sigma \max_i W_i\Big) \\ &= P\Big(Z + \Delta > \max_i W_i\Big) \end{aligned}$$

where $\Delta = \delta/\sigma$. Conditioning on $Z = x$:

$$\begin{aligned} P(\text{correct}) &= \int_{-\infty}^{\infty} P\Big(\max_i W_i < x + \Delta\Big)\, \phi(x)\, dx \\ &= \int_{-\infty}^{\infty} \prod_{i=1}^{M-1} P(W_i < x + \Delta)\, \phi(x)\, dx \\ &= \int_{-\infty}^{\infty} \Phi(x + \Delta)^{M-1}\, \phi(x)\, dx. \end{aligned}$$

This establishes (1). □ (Step 1)

Step 2: Gumbel asymptotics for the noise maximum

Lemma 2 (Extreme value concentration)**.** *Let $W_1, \ldots, W_{M-1} \overset{iid}{\sim} \mathcal{N}(0,1)$. Then*

$$\max_i W_i = \sqrt{2 \log M} - \frac{\log(4\pi \log M)}{2\sqrt{2 \log M}} + O_P\Big(\frac{1}{\sqrt{\log M}}\Big).$$

In particular, $\max_i W_i \to \sqrt{2 \log M}$ in probability as $M \to \infty$.

Proof. This is the classical Fisher–Tippett–Gnedenko theorem for the Gaussian case. The normalizing constants for the Gumbel limit of the Gaussian maximum are $a_M = \sqrt{2\log M} - \frac{\log\log M + \log(4\pi)}{2\sqrt{2\log M}}$ and $b_M = 1/\sqrt{2\log M}$. The random variable $(\max_i W_i - a_M)/b_M$ converges in distribution to a Gumbel random variable with mean $\gamma \approx 0.5772$ (Euler–Mascheroni constant). Since $b_M \to 0$, the maximum concentrates around $a_M \sim \sqrt{2\log M}$. □

Substituting $M = \bar{d}^h$:

$$\max_i W_i \approx \sqrt{2\log(\bar{d}^h)} = \sqrt{2h\log\bar{d}}. \tag{3}$$

Step 3: Phase transition

Retrieval succeeds when the signal exceeds the noise maximum:

$$S_a > \max_i S_{v_i} \iff Z + \Delta > \max_i W_i.$$

Since $Z = O_P(1)$ and $\max_i W_i \approx \sqrt{2h\log\bar{d}}$ (Lemma 2), the success event is determined by whether Δ exceeds the noise floor:

Case 1: $\Delta > \sqrt{2h\log\bar{d}}$. Then $Z + \Delta > \max_i W_i$ with high probability. Formally, for any $\varepsilon > 0$:

$$P(\text{correct}) \geq P(Z > -\varepsilon) \cdot P\left(\max_i W_i < \Delta - \varepsilon\right) \to 1$$

as $D \to \infty$ (since $\Delta - \varepsilon > \sqrt{2h\log\bar{d}}$ implies $P(\max_i W_i < \Delta - \varepsilon) \to 1$ by the Gumbel convergence).

Case 2: $\Delta < \sqrt{2h\log\bar{d}}$. Then $\max_i W_i > \Delta + \varepsilon$ with high probability for some $\varepsilon > 0$, and:

$$P(\text{correct}) \leq P(Z > \varepsilon) + P\left(\max_i W_i < \Delta + \varepsilon\right) \to 0 + 0 = 0.$$

Critical hop count. The transition occurs when $\Delta = \sqrt{2h^*\log\bar{d}}$. Substituting $\Delta = \delta\sqrt{D}$ and solving for h^*:

$$\delta\sqrt{D} = \sqrt{2h^*\log\bar{d}} \implies \delta^2 D = 2h^*\log\bar{d} \implies h^* = \frac{\delta^2 D}{2\log\bar{d}}.$$

□ (Theorem 1)

5 Corollaries

Corollary 3 (Density dependence)**.** *For fixed signal margin δ and embedding dimension D, the critical hop count h^* is inversely proportional to $\log\bar{d}$:*

$$h^*(\bar{d}_1)/h^*(\bar{d}_2) = \log\bar{d}_2/\log\bar{d}_1.$$

Denser graphs have lower h^* and therefore more severe cliffs.

Proof. Immediate from (2). □

For our two datasets: $h^*_{\text{PrimeKG}}/h^*_{\text{Amazon}} = \log(9.1)/\log(62.6) = 2.21/4.14 = 0.53$. That is, PrimeKG's critical hop is roughly half of Amazon's, consistent with PrimeKG's cliff appearing at $h \approx 2$ and Amazon's degradation being gradual through $h \approx 4$–5.

Corollary 4 (Sharpness of the transition)**.** *On a graph with mean degree $\bar{d}$, the noise floor increases by $\sqrt{\log \bar{d}}$ per hop increment. For $\bar{d} \gg 1$, this exceeds the $O(1)$ fluctuations of Z, so the transition from $P(\text{correct}) \approx 1$ to $P(\text{correct}) \approx 0$ occurs within $O(1/\log \bar{d})$ hops. Dense graphs exhibit sharper cliffs.*

Proof. The noise floor at hop h is $\sqrt{2h \log \bar{d}}$ and at hop $h+1$ is $\sqrt{2(h+1)\log \bar{d}}$. Near $h = h^*$, the increment is approximately:

$$\sqrt{2(h^*+1)\log \bar{d}} - \sqrt{2h^* \log \bar{d}} \approx \frac{\sqrt{2 \log \bar{d}}}{2\sqrt{2h^* \log \bar{d}}} = \frac{1}{2\sqrt{h^*}} = \frac{\sqrt{\log \bar{d}}}{\delta \sqrt{D}}.$$

This increment exceeds the standard deviation of Z (which is 1) when $\log \bar{d} > \delta^2 D$, which is precisely the regime $h^* < 1$—i.e., when the cliff occurs before even $h = 1$. For practical values ($h^* \geq 1$), the increment is $O(1/\sqrt{h^*})$, and the transition width is $O(\sqrt{h^*})$ hops. Since h^* is small for dense graphs, the transition is sharp. □

Corollary 5 (Independence from retrieval algorithm)**.** *The bound in Theorem 1 applies to* any *retrieval algorithm that selects candidates based on embedding similarity in $\mathbb{R}^D$, including beam search, iterative methods, and adaptive routing. The phase transition at h^* is a property of the embedding geometry and graph density, not of the retrieval algorithm.*

Proof. Any embedding-based retrieval algorithm must ultimately rank candidates by some function of their embeddings. The noise maximum $\max_i W_i$ is a property of the candidate set, not the algorithm. No algorithm can make the answer's similarity exceed the noise maximum when $\Delta < \sqrt{2h \log \bar{d}}$, because this is a property of the similarity scores themselves. □

Remark 2 (Why beam search and adaptive routing fail)**.** *Corollary 5 explains our empirical finding that beam search and adaptive routing do not overcome the cliff. Beam search prunes candidates at each hop using embedding similarity—but the same similarity scores that cannot distinguish the answer from noise at $h > h^*$ also cannot guide the beam toward the answer. Adaptive routing selects among configurations that all use the same embedding similarity as their ranking function. Neither approach changes Δ, D, or $\bar{d}$, so neither can move h^*.*

Remark 3 (How to push h^* higher)**.** *Equation* (2) *identifies three levers for increasing h^*:*

1. ***Increase** δ (signal margin): Use better embeddings or fine-tune on the KG. Relation-type-aware traversal (e.g., Think-on-Graph) effectively increases δ by filtering irrelevant candidates, reducing M without reducing signal.*

2. ***Increase** D (embedding dimension): Higher-dimensional embeddings can distinguish more candidates, but with diminishing returns ($h^* \propto D$, and compute/memory scale with D).*

3. ***Decrease** $\bar{d}$ (effective degree): Edge-type filtering reduces the effective degree from $\bar{d}$ to $\bar{d}_{\text{eff}} \ll \bar{d}$, dramatically increasing h^*. This is why MoR's planning stage and relation-type routing are effective.*